

Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data*

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Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

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dataset covers more than 280,000 U.S. firms, more than 160 million employment spells from roughly 65 million unique workers between 2015 and 2025, and almost 200 million job postings. A key advantage of the data is the standardized seniority classification assigned to each position by Revelio’s algorithm, which enables us to track junior and senior employment within firms over time.⁴

We identify GenAI adoption by detecting job postings that explicitly recruit “GenAI integrator” roles. The method follows the approach of [Hampole et al. \(2025\)](#) and proceeds in two steps: first flagging postings with GenAI-related keywords, then using a large language model (LLM) to determine whether the posting reflects a genuine integrator position—one dedicated to implementing or operating GenAI technology in the firm’s workflow. A firm is classified as an adopter if it has posted at least one such vacancy, thereby capturing firms that have actively initiated the integration of GenAI into their operations.⁵

By this definition, 10,433 firms in our sample adopted GenAI by March 2025. Although adopters represent only about 3.7 percent of the 281,000 firms in our sample, they are substantially larger on average and account for roughly 16 percent of total employment in our dataset. Our analysis shows that adoption of GenAI was minimal and relatively stable prior to 2023, but accelerated sharply thereafter, with a surge of new firms posting integrator roles following the release of ChatGPT in late 2022. We further validate this firm-level adoption measure using worker-side data. Applying a list of GenAI keywords to position descriptions, we find a sharp differential increase in AI-related mentions at adopting firms after the launch of ChatGPT, supporting the interpretation that our adoption measure captures genuine changes in workplace technology use.

The main empirical exercise of the paper is to test the framework’s central implication: whether GenAI adoption disproportionately reduces junior employment relative

⁴We validate the data in Appendices [A.1](#) and [A.2](#). First, we show that the composition of the LinkedIn sample relative to QCEW, OEWS, and QWI benchmarks remains stable over the GenAI diffusion period. Second, we test for differential reclassification: the concern that adopting firms changed how positions were classified as junior after 2022. Using both job-title keywords and workers’ actual labor-market experience, we find that the content of each seniority group evolves similarly at adopting and non-adopting firms, with no break around the ChatGPT launch. This indicates that the decline in junior employment is unlikely to be driven by changes in seniority classification.

⁵Our concept of a GenAI integrator is closely related to the “robot integrator” in [Acemoglu and Restrepo \(2020\)](#), defined as “companies that install, program, and maintain robots.” They use the presence of robot integrators as a proxy for local robot adoption and show that commuting zones more exposed to robots have a higher number of integrators.